Year 2 Assessed Problems

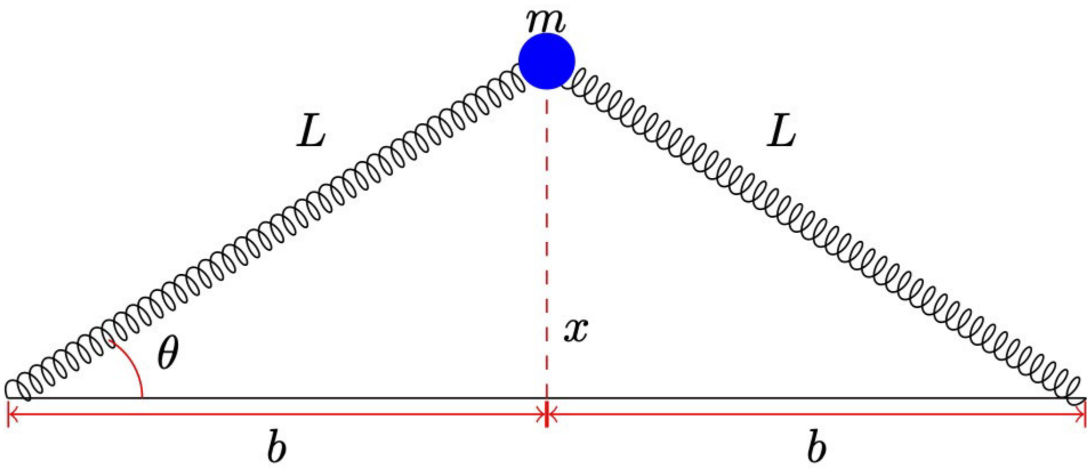
Semester 2

Assessed Problems

SOLUTIONS TO BE SUBMITTED
ON CANVAS BY

Wednesday 18 March

at 17:00



1. The diagram shows a mass m attached to two springs of force constant k.Each spring has a natural length a, and in equilibrium when the mass is at rest each spring is stretched to a length L. The mass moves in a horizontal plane on a frictionless table

   (a) Find the value of x that corresponds to the equilibrium value of the potential. And evaluate the potential to second order in x around this point. (2)

   (b) Evaluate the Euler-Lagrange equations or otherwise to demonstrate that when the system is displaced sideways through the small distance x as shown, the system exhibits simple harmonic motion. (6)

   (c) Find the angular frequency of oscillation. (2)

Y2 Neutrino Physics
Assessed problem sheet 3 (weeks 6–7)

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1. Consider an isotropic neutrino source located at the south pole of the Earth. Assume the Earth to be spherical with a radius of 6371 km.

a) Compute the distance from the source to a detector located near Birmingham (52nd parallel north). Compute the time is takes a high-energy neutrino to travel from the source to the detector. [**1**]

b) The detector described above aims at a study of neutrino oscillations driven by a mass splitting of $\Delta m^2 = 2 \times 10^{-3}$ eV^2. Assuming monochromatic neutrinos and two-flavour oscillations in vacuum, compute the neutrino energies (in GeV) required to observe the first and second oscillation maxima at the detector. [**1**]

c) The detector described above measures a survival probability at the oscillation maximum of 75%. What is the mixing angle (in degrees)? [**1**]

d) If the second oscillation maximum is observed at the detector near Birmingham, then the first maximum can be in principle observed in other places on the Earth surface. At which latitudes is it observable? [**1**]

2. A *threshold Cherenkov counter* can distinguish two types of charged particles by the fact that particles of one type emit Cherenkov radiation, while particles of the other type do not. Consider a nitrogen-filled threshold Cherenkov counter at room temperature and atmospheric pressure (refractive index $n = 1.000298$) aiming to distinguish pions (π^+) from kaons (K^+) in a monochromatic beam. In which momentum range can this counter operate? [**1**]

Assessed problem sheet 3

1. The semi-empirical mass formula for the binding energy is:

$$B = a_\mathrm{v} A - a_\mathrm{s} A^{2/3} - a_\mathrm{c} \frac{Z(Z-1)}{A^{1/3}} - a_\mathrm{sym} \frac{(Z-N)^2}{A} + a_p \frac{\delta(Z,N)}{A^{1/2}}$$

with the coefficients

$a_v = 15.8\,\mathrm{MeV}$, $a_s = 18.3\,\mathrm{MeV}$, $a_c = 0.714\,\mathrm{MeV}$, $a_{sym} = 23.2\,\mathrm{MeV}$, $a_p = 12\,\mathrm{MeV}$

(a) Calculate the binding energies for $^{208}_{82}\mathrm{Pb}$ and for $^{204}_{80}\mathrm{Hg}$. **[2]**

(b) Using your results from (a) and the measured binding energy for $^{4}_{2}\mathrm{He}$ of $B(^{4}_{2}\mathrm{He}) = 28.296\,\mathrm{MeV}$, calculate the Q-value for the α-decay of $^{208}_{82}\mathrm{Pb}$ and show that you would expect this decay to happen spontaneously. **[1]**

(c) The actual Q-value is only $517\,\mathrm{keV}$, and the decay has never been observed. Why is it not surprising that $^{208}_{82}\mathrm{Pb}$ is much more strongly bound than the formula suggests? **[1]**

(d) Give one experimental observation that suggests that the nucleon-nucleon force is spin or angular momentum dependent. **[1]**

Assessed Problems 4

Richard Mason

Problem 1. Consider the Schrödinger equation in 2 dimensions. We consider a circularly symmetric potential which has the form

$$V(x,y) = -\frac{\lambda}{x^2+y^2}$$

1. Convert the Schrödinger equation in two dimensional Cartesian coordinates with the above potential into plane polar coordinate using $x = r\cos\theta$ and $y = r\sin\theta$ and show it is given by **[4]**

$$-\frac{\hbar^2}{2m}\left[\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}\right]\psi - \frac{\lambda}{r^2}\psi = E\psi$$

   You will need that the Laplacian in two dimensions for any curvilinear coordinate system, e_1, e_2 is given by

$$\nabla^2 = \frac{1}{H}\left(\frac{\partial}{\partial e_1}\left[\frac{H}{h_1^2}\frac{\partial}{\partial e_1}\right] + \frac{\partial}{\partial e_2}\left[\frac{H}{h_2^2}\frac{\partial}{\partial e_2}\right]\right)$$

   where $H = h_1 h_2$ and h_i are the scale factors defined by (in two dimensions)

$$h_i = \sqrt{\left(\frac{\partial x}{\partial e_i}\right)^2 + \left(\frac{\partial y}{\partial e_i}\right)^2}$$

2. Employ separation of variables to find the radial and angular equations. **[2]**

3. Solve the angular equations, and provide a quantisation on the separation constant which ensures single valuedness. **[1]**

4. The n^{th} Bessel function satisfies the ODE

$$x^2\frac{d^2y_n}{dx^2} + x\frac{dy_n}{dx} - (n^2 - x^2)y_n(x) = 0$$

   Write the solution to the radial equation in terms of $y_n(x)$. You may want to define

$$\mu = \frac{2m\lambda}{\hbar^2} \qquad \epsilon = \frac{2mE}{\hbar^2} \qquad \beta = m^2 - \mu$$

   [3]

Year 2: Observational Astronomy

Assessed Problem Sheet #3

1. An astronomer wishes to resolve a binary star with a separation of 0.8 arcsec, using a filter centered at 500nm. Ignoring the effects of the atmosphere, calculate the minimum size of telescope needed to resolve the binary. [Hint: base your answer on the size of the Airy disk.] **[2 marks]**

2. The seeing is typically measured to be $\mathrm{FWHM} = 0.7\mathrm{arcsec}$ at $\lambda = 2.2\mathrm{um}$ (i.e. in the near-infrared) at the site on which the telescope is located. Will the astronomer be able to resolve the binary star in an observation at 500nm with a telescope of diameter 1 metre from this site? Justify your answer, **[4 marks]**

3. The same astronomer would like to observe the Pleiades star cluster with the 1 metre telescope from part 2. Plaiedes subtends 2 degrees on the sky. A camera that contains CCDs that cover a square $10 \times 10\mathrm{cm}$ area of the focal plane at the f/2 prime focus of this telescope is available. Calculate whether the astronomer can fit Pleiades within a single observation with this camera. **[4 marks]**